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Machine learning for neural & biological systems — rigorous measurement of representations, from cortical populations to single-cell atlases.

NeurIPS 2026 · Sydney, 6–13 Dec — NeurReps poster #46, Fri 11 Dec, ICC Sydney Hall 4 · Open to research roles & collaborations in AI × life science

I lead AI strategy and newsroom AI systems at the SCMP (RAG copilots, AI proofreading, timeline automation) and run an independent research programme on how representations should be measured before they are believed. Previously R&D in Nokia, Microsoft. Moving toward independent research and advisory work at the AI × life-science interface.

1 · NEURREPS 2026 — EXTENDED ABSTRACT #46 (THIS POSTER)

## Three guards for subspace claims: chance, stability and metric

Question. Do evoked neural population patterns live inside the spontaneous-activity subspace? Classics said yes; modern large-scale recordings say near-orthogonal. The field's standard "containment" number is not one well-posed quantity.

Three guards. Report excess over an explicit rotation null — a random k-dim subspace already captures k/n of any probe (a 10–30-point floor at classic scales). Screen conditioning by the boundary gap $\sigma_k/\sigma_{k+1}$ — median gaps fall below 1.05 beyond k  3–5, while published work uses k = 10–128. Declare the metric — rate normalization moves hippocampal cross-state excess from 0.32–0.61 to 0.02–0.19.

Result. The literature's disagreement decomposes into chance-floor inflation, ill-conditioned projectors and undeclared metric + state mixing; the small V1 excess survives all guards. Data: 60 Neuropixels populations, ~10k-neuron two-photon V1, monkey DMFC, HD thalamus, hippocampal CA1; measures ship with a 15-test known-answer battery.

2 · RECOMB 2027 — SUBMITTED (EASYCHAIR #22)

## Separating higher-order information from prediction in cancer co-mutation

Most "higher-order" claims in biology rest on pairwise statistics. This work asks how much of cancer co-mutation structure is genuinely beyond pairs.

It separates two quantities that are often conflated: the higher-order information present in the data versus the predictive gain a model can actually extract from it.

Solo-authored; code at github.com/freemanhui/beyond-pairs.

3 · C. ELEGANS — SINGLE-CELL PROGRAMME

## A worm virtual cell: atlas, state models & a small-RNA study

C. elegans is the only animal with a complete cell lineage and a complete connectome (302 neurons) — a whole-organism testbed the large labs leave alone.

Atlas direction: a worm-specific embedding and an open cell-type annotation tool on Arc scBaseCount data (295,473 cells, 58 samples), with a daf-2 ageing-perturbation case study under strict held-out validation — baselines first, per the Virtual Cell Challenge lesson.

Companion paper: "The inheritance state reweights the small RNA pool beyond uniform gain in C. elegans" — manuscript in submission at Briefings in Functional Genomics (Oxford University Press).

**METHODS TOOLKIT**

Representation geometry · information theory · topological data analysis (persistent homology) · single-cell ML (scVI, transformers) · large-scale LLM experimentation · evaluation design — nulls, conditioning, metric audits

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